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Gunneng

(54) A FASTENING ASSEMBLY FOR USE IN A CARRIER SUCH AS A PICK-UP/TRUCK BED AND A METHOD FOR DETACHABLY FASTENING AN OBJECT TO A CARRIER SUCH AS A PICK-UP/TRUCK BED

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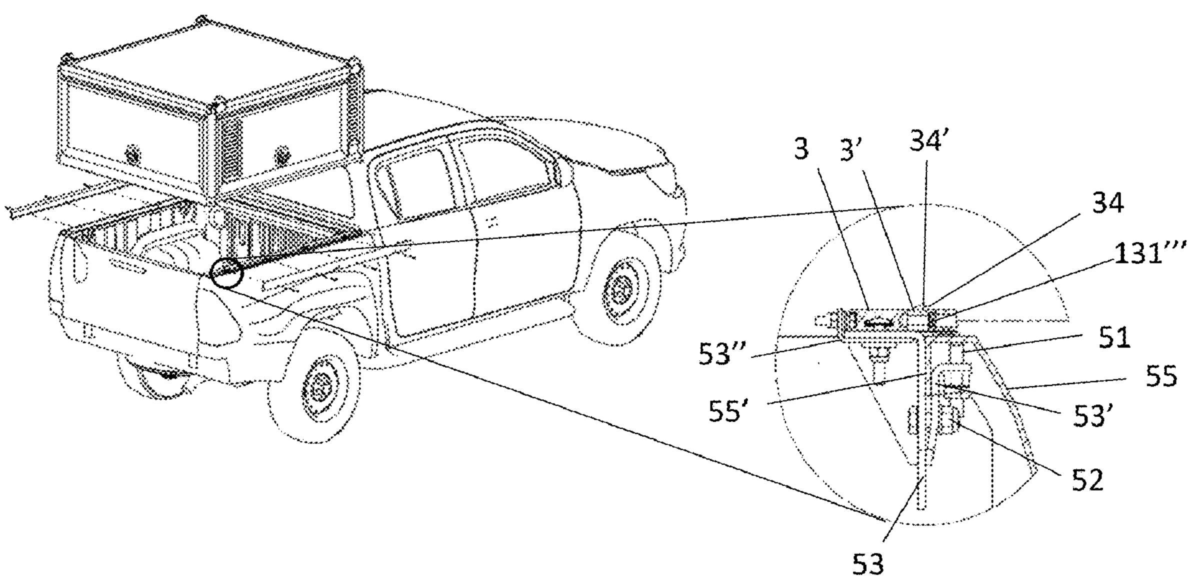
(57) **ABSTRACT**

The disclosure relates to a fastening assembly for use in a carrier such as a pick-up/truck bed comprising: one or more casing profile lists attached to an object to be detachably fastened, the casing profile list is arranged at a side of the object facing a casing profile locking assembly arranged on the carrier, the casing profile locking assembly comprising: a horizontal bottom side profile list, one or more of a first fastener for holding the horizontal bottom side profile list, wherein the horizontal bottom side profile list fastened to the carrier, and arranged to receive the casing profile list, and a vertical bottom side profile list arranged to interact with the horizontal bottom side profile list for fastening the casing profile list in a locking grip. The disclosure further relates to a method for fastening an object to a carrier such as a pick-up/truck bed the method.



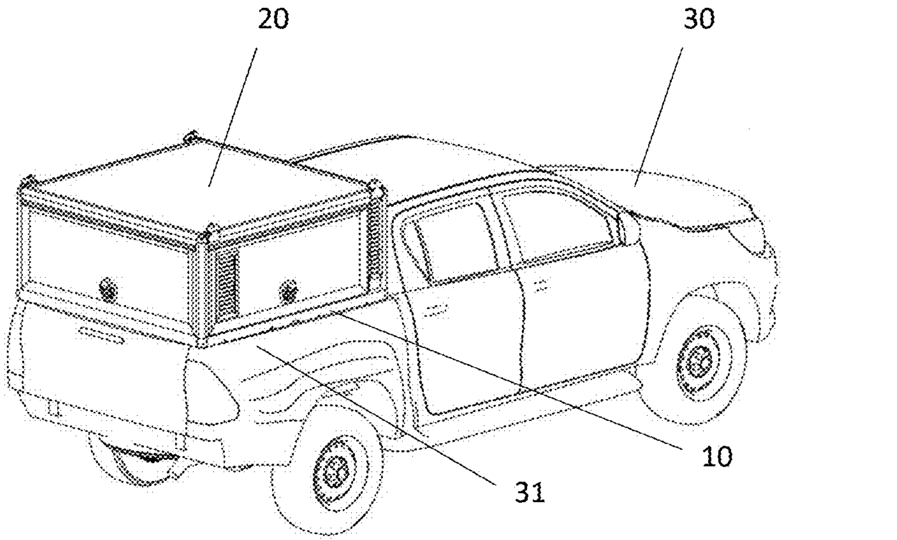


Fig. 1

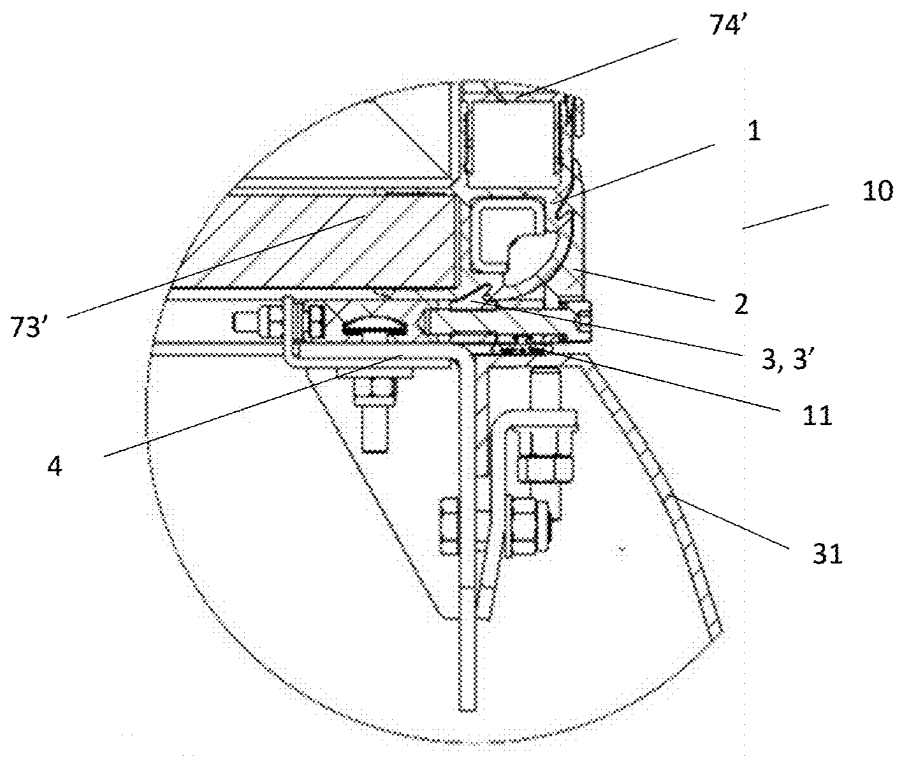


Fig. 2

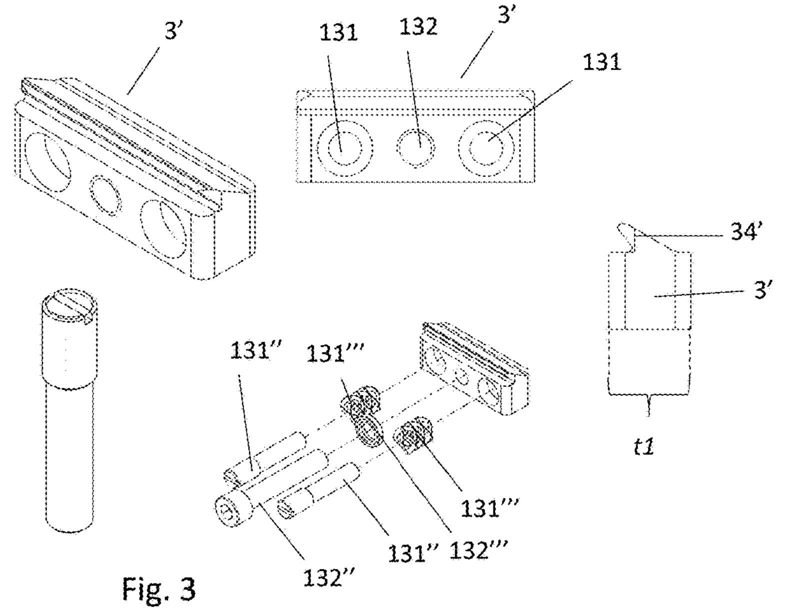


Fig. 3

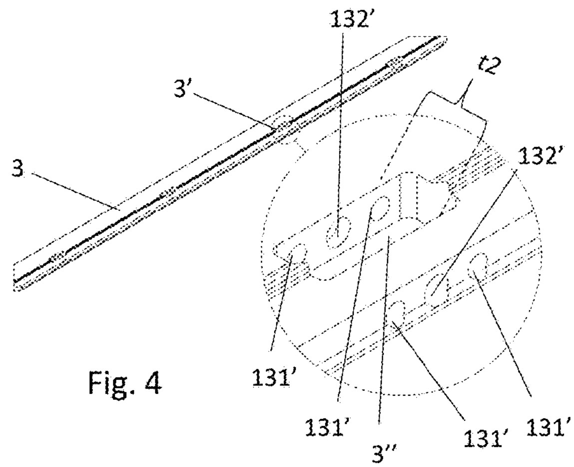


Fig. 4

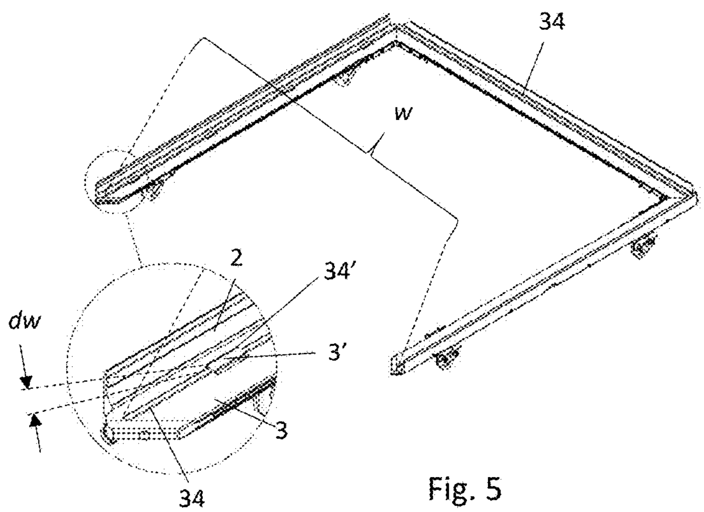


Fig. 5

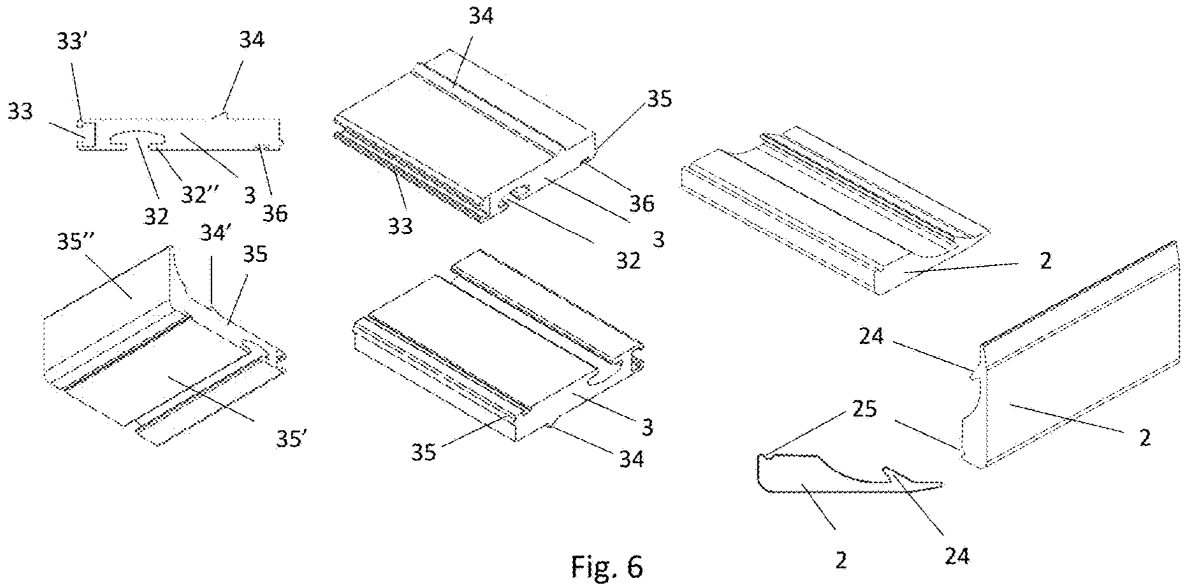


Fig. 6

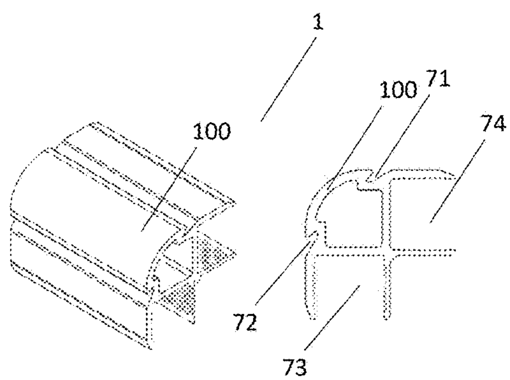


Fig. 7

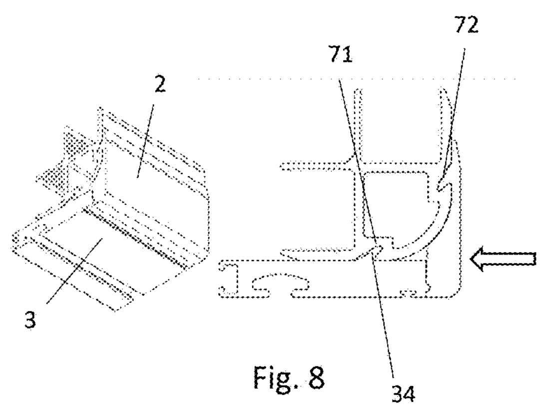


Fig. 8

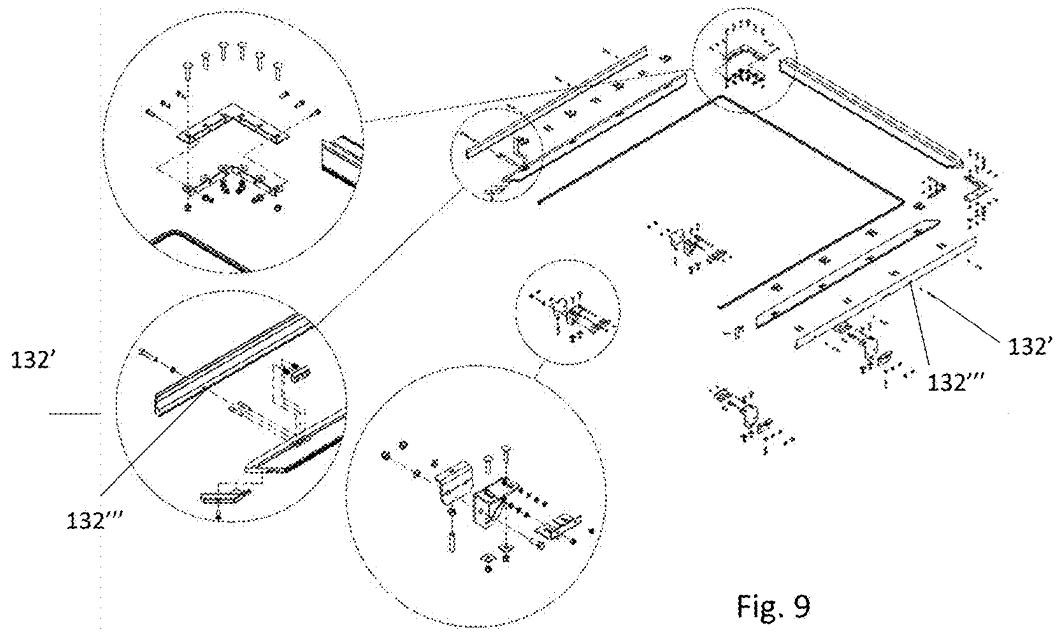


Fig. 9

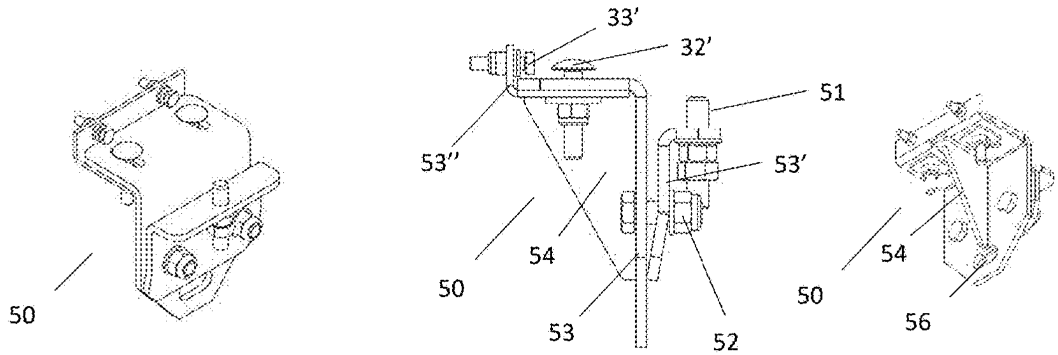


Fig. 10

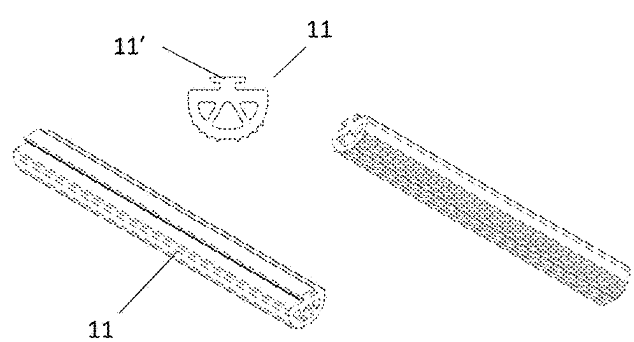


Fig. 11

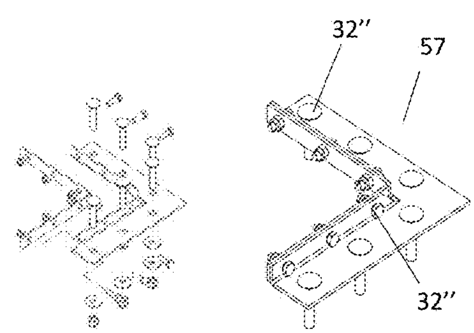


Fig. 12

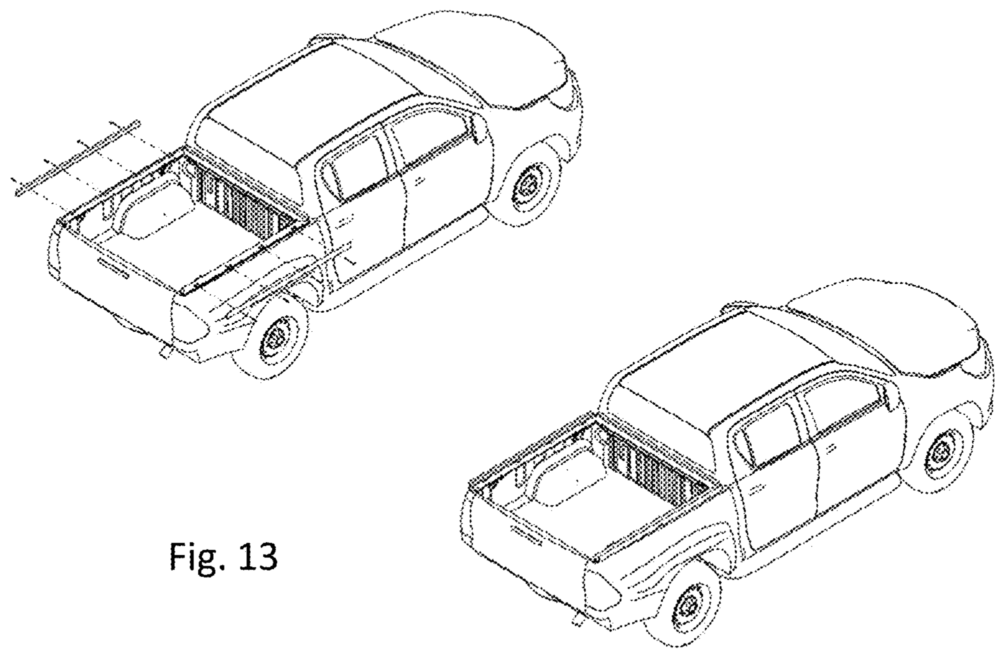


Fig. 13

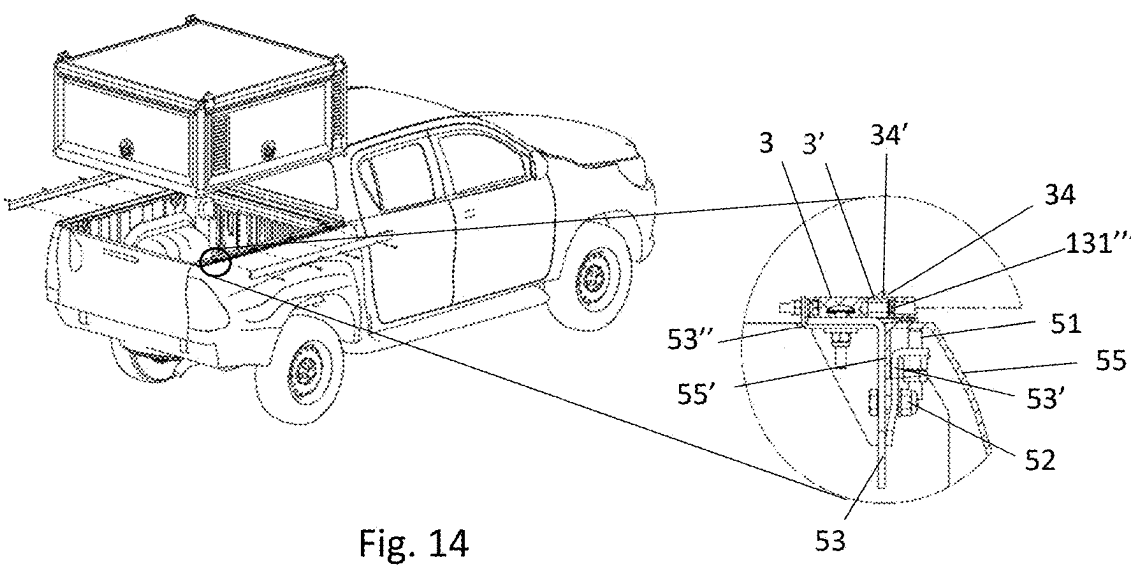


Fig. 14

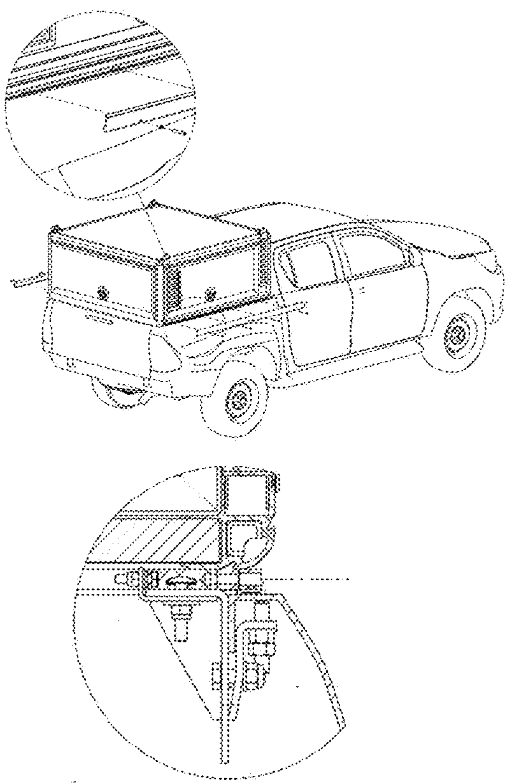


Fig. 15

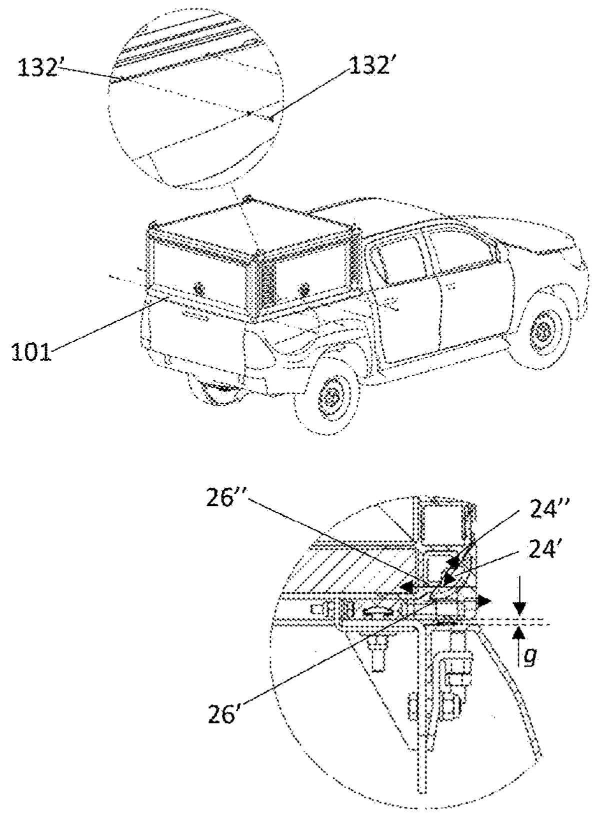


Fig. 16

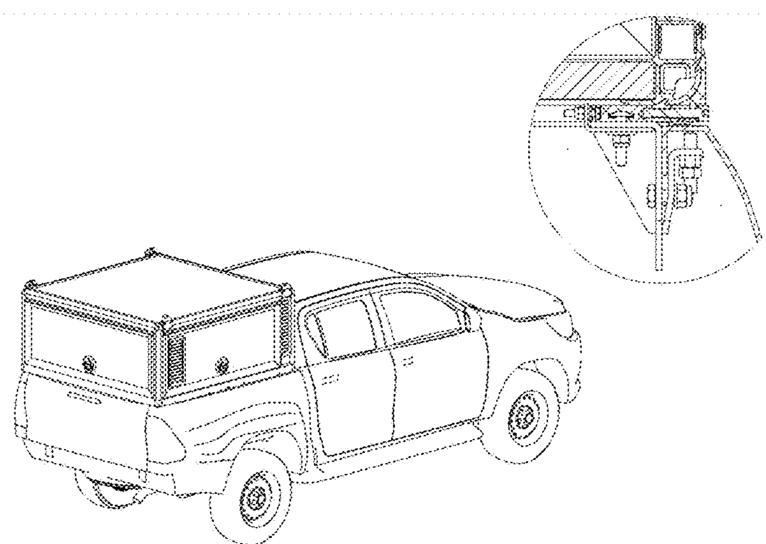


Fig. 17

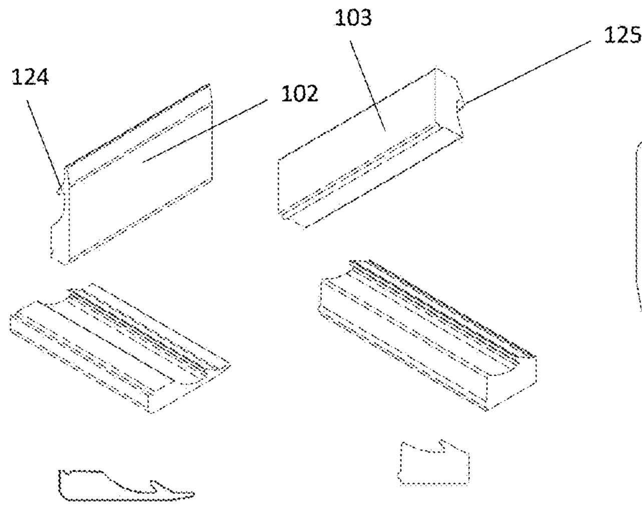


Fig. 18

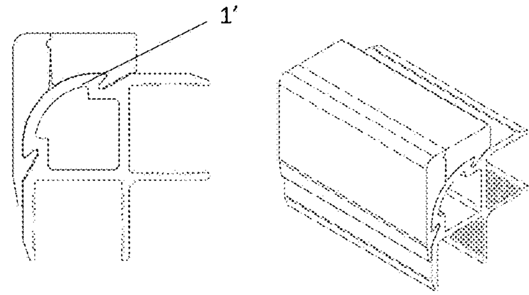


Fig. 19

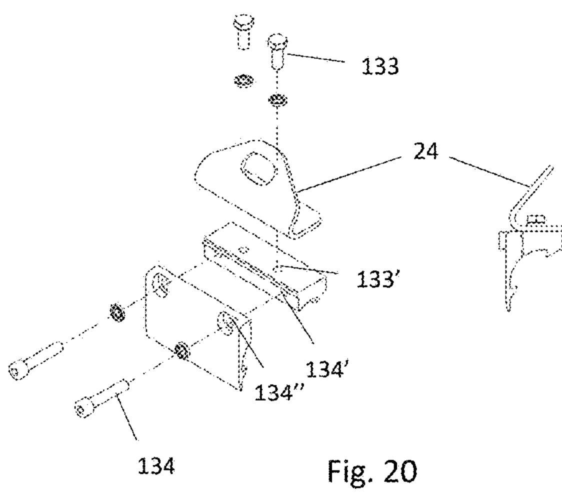


Fig. 20

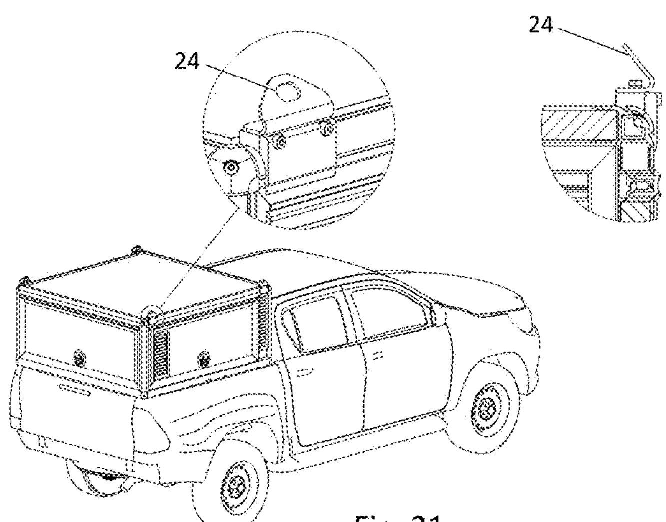


Fig. 21

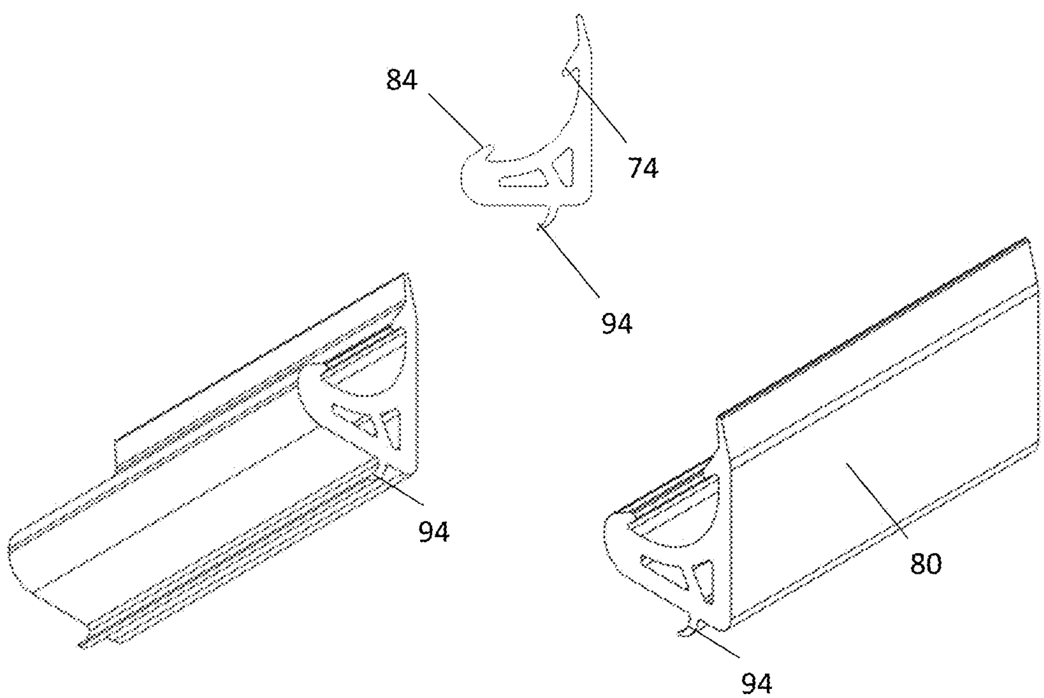


Fig. 22

A FASTENING ASSEMBLY FOR USE IN A CARRIER SUCH AS A PICK-UP/TRUCK BED AND A METHOD FOR DETACHABLY FASTENING AN OBJECT TO A CARRIER SUCH AS A PICK-UP/TRUCK BED

TECHNICAL FIELD

[0001] The present disclosure relates to a fastening assembly for use in a carrier such as a pick-up/truck bed and a method for fastening an object to a carrier such as a pick-up/truck bed the method. More specifically, the disclosure relates to a fastening assembly for use in a carrier such as a pick-up/truck bed and a method for fastening an object to a carrier such as a pick-up/truck bed the method as defined in the introductory parts of the independent claims.

BACKGROUND ART

[0002] A problem with the solutions of the prior art is that custom built carrier equipment needs to be built to accommodate to regulations when fastening all covering boxes, crates, hoods or the like onto a pick-up/truck bed. Sometimes it is sufficient to customize these to adapt to cargo lashing belt connectors attached to lashing points/eyes suitable for the purpose. When the requirements to attach objects that must be attached above the side panels for any reason, there are huge challenges to comply with safety regulations. In order to attach the objects safely this has previously been solved by customizing the carrier at huge costs and individual adaptations to each version of carrier used. There is thus a need for improved fastening assembly for use in a carrier, such as a pick-up/truck, for attaching an object to the carrier.

SUMMARY

[0003] It is an object of the present disclosure to mitigate, alleviate or eliminate one or more of the above-identified deficiencies and disadvantages in the prior art and solve at least the above mentioned problem. According to a first aspect there is provided a fastening assembly for use in a carrier such as a pick-up/truck bed comprising: one or more casing profile lists for being attached to an object to be detachably fastened, the casing profile list is arranged at a side of the object facing a casing profile locking assembly arranged on the carrier, the casing profile locking assembly comprising: a horizontal bottom side profile list, one or more of a first fastening means for holding the horizontal bottom side profile list, wherein the horizontal bottom side profile list being fastened to the carrier, and being arranged to receive the casing profile list, and a vertical bottom side profile list for being arranged to interact with the horizontal bottom side profile list for fastening the casing profile list in a locking grip.

[0004] The advantage is that it is now facilitated a common ("one size" fits all type) method, that may be non-intrusive on the carrier bodywork/framework, for providing a quick, safe, flexible fastening of objects to a carrier. Typically a carrier is pick-up/truck, and the object may be a large crate/box to be arranged on the pick-up/truck bed above the side panels. As long as the object is provided with casing profile list, or having casing profile list arranged to it, the object itself provides a portion of the fastening assembly.

[0005] According to some embodiments, the casing profile list may comprise a first connecting element, and the horizontal bottom side profile list comprise a first corresponding connecting element for interacting in a guiding operation with the first connecting element.

[0006] Thus it is advantageously provided an easy way to guide the loading of the object in a quick and safe manner.

[0007] According to some embodiments, the casing profile list may further comprise a second connecting element, and the vertical bottom side profile list comprise a second corresponding connecting element for interacting in a locking grip with the second connecting element.

[0008] The advantage of the latter element is that the "locking" of the fastening assembly is done post arranging the object on the carrier at convenient and easy locations along the side of the carrier.

[0009] According to some embodiments, the horizontal bottom side profile list and the vertical bottom side profile list is provided in a combined bottom side profile list.

[0010] For non-easy to reach sides after object is arranged on the carrier, for example the front side of a box reaching all the way to the front panel which often is located next to a drivers cabin back wall, it is provided a combined profile list preventing forward gliding of the object.

[0011] According to some embodiments, the fastening assembly may further comprising: two or more locking member assemblies comprising a lock member connecting element corresponding to the first corresponding connecting element of the horizontal bottom side profile list, and the horizontal bottom side profile list comprising two or more grooves for receiving the locking members, and the locking member assembly having: at least one locking member holding means for holding the locking members in the horizontal bottom side profile list groove, and at least one locking member receiving element, the horizontal bottom side profile list, the locking member receiving element, and the vertical bottom side profile list having corresponding conduits for receiving a connecting bolt for fastening the vertical bottom side profile list to the horizontal bottom side profile list.

[0012] Thus, a locking element having a resilient lock-in/lock-out performance facilitating easy attachment and release is provided.

[0013] According to some embodiments, the first fastening means may comprising: a vertical downward pulling fastening element for holding the horizontal bottom side profile list in the vertical direction to the carrier, and a horizontal working fastening element for aligning and holding the horizontal bottom side profile list in the horizontal direction relative the carrier.

[0014] Thus, a separate assembly is arranged to the carrier pre-arranging the object in a secure manner.

[0015] According to some embodiments, the vertical downward pulling fastening element may be comprised of: a vertical holding element being arranged in a second fastening means, and a first receiving groove being arranged longitudinally in the underside of the horizontal bottom side profile list, the first receiving groove having a form to receive and hold the vertical holding element.

[0016] According to some embodiments, the horizontal working fastening element may be comprised of: a horizontal holding element being arranged in a second fastening means, and a second receiving groove being arranged longitudinally in a first side of the horizontal bottom side profile list, the second receiving groove having a form to receive and hold the horizontal holding element.

[0017] Thus, a quick and reliable way to arrange the horizontal holding element is provided pre-arranging the object to the carrier.

[0018] According to some embodiments, the horizontal bottom side profile list may further comprise a first weight transfer element, and the vertical bottom side profile list further comprising a second weight transfer element, wherein the first weight transfer/alignment element and the second weight transfer/alignment element cooperates to transfer and hold the weight of the object attached to the carrier.

[0019] According to some embodiments, the second fastening means comprise: the vertical holding element, the horizontal holding element, and a carrier fastening element for fastening the second fastening means to the carrier.

[0020] Combining a fastening module with the object fastening assemblies provides a further flexibility to the assemblies, and easy fitting in a non-intrusive manner may be provided.

[0021] According to some embodiments, the carrier may be a pick-up truck/lorry bed having the carrier fastening elements for fastening the second fastening means to the carrier arranged on any of the top side of side and front bed panel, the topside of the panel having an inward facing collar to provide an upside down oriented fold, and the carrier fastening elements the fastening comprises: a lower vertical portion of an inner element for being arranged on the inside of the upside down oriented fold, a lower vertical portion of an outer element or being arranged on the outside of the upside down oriented fold, and a fastening device arranged to pinch and hold the upside down oriented fold between the lower vertical portion of an inner element and the lower vertical portion of an outer element, the inner element further comprising: an upper horizontal portion for holding the vertical holding element, and an upper extension portion for holding the horizontal holding element, and the outer element further comprising: an upper horizontal portion for holding a stabilizing bolt.

[0022] This embodiment will fit to many Pick-up trucks on the market, but other mechanism for fastening the second fastening elements may be provided.

[0023] According to some embodiments, the second fastening means may further comprise: a third fastening means for further securing the second fastening means to the carrier.

[0024] This embodiment will fit to many Pick-up trucks on the market, but other mechanism for fastening the second fastening elements may be provided.

[0025] According to some embodiments, the third fastening means comprise a strap fastening element providing fastening of a securing connector/strap to the carrier.

[0026] It may be advantageous to further secure any objects with easy accessible strap fastening elements.

[0027] According to some embodiments, the carrier may be a pick-up truck/lorry bed having a horizontal bottom side profile list arranged on the top side of side and front bed panel, and the object is a container having mounted: on its underside two longitudinal parallel running casing profile list for being cooperating with the horizontal bottom side profile list arranged on the side panel, and on the forward facing underside a further profile list for being cooperating with the horizontal bottom side profile list arranged on the front panel, and the second fastening means are arranged on the top side of side bed panel of the pick-up truck/lorry bed.

[0028] According to some embodiments the first connecting element, and the first corresponding connecting element of the horizontal bottom side profile list is composed of a protrusion interacting with a groove.

[0029] According to some embodiments the second connecting element, and the second corresponding connecting element of the vertical bottom side profile list is composed of a protrusion interacting with a groove.

[0030] These protrusions and grooves correspond in form, and may be interchanged. In some embodiments the first corresponding connecting element allow for receiving the object in a straight downward movement loading operation.

[0031] According to some embodiments, the fastening assembly may further comprise a roof casing profile list for being attached to an object to be detachably fastened, the roof casing profile list is arranged at a different side of the object facing the casing profile locking assembly arranged on the carrier, a horizontal top side profile list, and a vertical top side profile list for being arranged to interact with the horizontal top side profile list for fastening the roof casing profile list in a locking grip.

[0032] According to some embodiments, the fastening comprises: a fourth fastening means for securing a further object to the roof casing profile list.

[0033] The roof casing profile list and its associated parts may be arranged to an object independently of the fastening assembly.

[0034] By modifying the fastening assembly slightly it may be possible to fasten additional rooftop objects to the object via this roof casing profile list.

[0035] According to some embodiments, the fastening assembly may comprise: a blind casing profile list for being attached to an object to be detachably fastened, the blind casing profile list is arranged at the lower end of the rear side of the object facing the upper end of a tailgate, and a tail gasket having an inner form mirroring and adapted to being arranged to interact with the first and second connecting elements of the blind casing profile list in a locking manner, and the tail gasket may further comprise a downward pointing flange portion being arranged to close a gap between the tail gasket and the top of the tailgate.

[0036] For any side not being locked by the fastening assembly, it is advantageously provided a seal for preventing dirt to enter into the space below the object.

[0037] According to a second aspect there is provided a method for fastening an object to a carrier such as a pick-up/truck bed the method comprising the steps: arranging one or more casing profile list according to the first aspect to an object to be arranged on a carrier, mounting a two or more carrier fastening element to the carrier, mounting one or more horizontal bottom side profile lists to the two or more carrier fastening element, mounting the casing profile list arranged on the object into the corresponding horizontal bottom side profile lists, mounting one or more vertical bottom side profile list to the horizontal bottom side profile lists, and thereby fastening the object in a tight grip.

[0038] According to some embodiments, the method comprises the further steps: arranging a blind casing profile list at the lower end of the rear side of the object facing the upper end of a tailgate of the carrier.

[0039] According to some embodiments, the method comprises the further steps; fastening one or more roof casing profile list to the upper portion of an object at the opposite side of the object facing the casing profile locking assembly arranged on the carrier, and arranging a fourth fastening means for securing a further object to the roof casing profile list.

[0040] Effects and features of the second aspect are to a large extent analogous to those described above in connection with the first aspect. Embodiments mentioned in relation to the first aspect are largely compatible with the second aspect.

[0041] The present disclosure will become apparent from the detailed description given below. The detailed description and specific examples disclose preferred embodiments of the disclosure by way of illustration only. Those skilled in the art understand from guidance in the detailed description that changes and modifications may be made within the scope of the disclosure.

[0042] Hence, it is to be understood that the herein disclosed disclosure is not limited to the particular component parts of the device described or steps of the methods described since such device and method may vary. It is also to be understood that the terminology used herein is for purpose of describing particular embodiments only, and is not intended to be limiting. It should be noted that, as used in the specification and the appended claim, the articles "a", "an", "the", and "said" are intended to mean that there are one or more of the elements unless the context explicitly dictates otherwise. Thus, for example, reference to "a unit" or "the unit" may include several devices, and the like. Furthermore, the words "comprising", "including", "containing" and similar wordings do not exclude other elements or steps.

[0043] The term "horizontal" and "vertical" is to be interpreted as the element's relative arrangement to each other, and wording is selected to fit the elements posture in the figures, but other embodiments are encompassed even if they may arrange the elements in other directions.

[0044] The term "carrier" is used to describe the entity that the fastening means of the present disclosure is used to arrange an object to, and shall be interpreted in widest meaning to at least comprise: pick-up trucks, lorries, barges, cars, boats, planes, drones or any type of carrier of fixed installation.

[0045] The terms "arranged on" and "onto" is to be understood not being connected to up or down or any direction. The phrases denotes how the parts relates to each other independent on word coordinates.

[0046] The terms "profile list" is to be understood as any type of profile list being manufactured from any type of material. Typically some profile lists are made from extruded aluminum, but other materials may advantageously be used adapted to the environment it is to be used and for the task to be solved.

BRIEF DESCRIPTIONS OF THE DRAWINGS

[0047] The above objects, as well as additional objects, features and advantages of the present disclosure, will be more fully appreciated by reference to the following illustrative and non-limiting detailed description of example embodiments of the present disclosure, when taken in conjunction with the accompanying drawings.

[0048] FIG. **1** shows a pick-up truck having an object/box arranged on the bed

[0049] FIG. **2** shows a cross section of the fastening means according to present disclosure

[0050] FIG. **3** shows an embodiment of the locking member assembly and parts

[0051] FIG. **4** shows an embodiment of the horizontal bottom side profile list and the groove for receiving the locking member

[0052] FIG. **5** shows an embodiment of the horizontal bottom side profile list arranged for receiving an object having a square/rectangular bottom form.

[0053] FIG. **6** shows an embodiment of the horizontal bottom side profile list and the vertical bottom side profile list seen from various angles

[0054] FIG. **7** shows an embodiment of a casing profile list seen from various angles

[0055] FIG. **8** shows an embodiment of shows a section of an embodiment of casing profile list with the horizontal bottom side profile list and the vertical bottom side profile list arranged to grip the casing profile list seen from various angles

[0056] FIG. **9** shows an embodiment of the fastening assembly in a disassembled manner, and with blown up sections

[0057] FIG. **10** shows an embodiment of the second fastening means to be arranged on the carrier

[0058] FIG. **11** shows an embodiment of the gasket for being arranged in the horizontal bottom side profile list

[0059] FIG. **12** shows an embodiment of a corner fastening profile

[0060] FIG. **13** shows an embodiment of a first step of arranging the fastening assembly on a pick-up truck

[0061] FIG. **14** shows an embodiment of a second step of arranging the fastening assembly on a pick-up truck

[0062] FIG. **15** shows an embodiment of a third step of arranging the fastening assembly on a pick-up truck

[0063] FIG. **16** shows an embodiment of a fourth step of arranging the fastening assembly on a pick-up truck

[0064] FIG. **17** shows an embodiment of a final step of arranging the fastening assembly on a pick-up truck

[0065] FIG. **18** shows an embodiment of shows a section of an embodiment of the horizontal top side profile list and the vertical top side profile list arranged to grip the casing profile list seen from various angles

[0066] FIG. **19** shows an embodiment of shows a section of an embodiment of the horizontal top side profile list and the vertical top side profile list gripping the casing profile list seen from various angles

[0067] FIG. **20** shows an embodiment of the elements of a fourth fastening means

[0068] FIG. **21** show how the fourth fastening means is arranged to provide fastening means for securing further objects to the roof casing profile list.

[0069] FIG. **22** shows an embodiment of a tail gasket

DETAILED DESCRIPTION

[0070] The present disclosure will now be described with reference to the accompanying drawings, in which preferred example embodiments of the disclosure are shown. The disclosure may, however, be embodied in other forms and should not be construed as limited to the herein disclosed embodiments. The disclosed embodiments are provided to fully convey the scope of the disclosure to the skilled person.

[0071] The figures exemplify embodiments of present disclosure arranging a box object on a pick-up truck, but the nature of the disclosure is such that it may be arranged to fasten any object to any carrier providing a suitable support for the fastening assembly as described. Thus, the reader should understand that the fastening assembly may be used on lorries, barges, cars, boats, planes, drones or any type of carrier of fixed installation.

[0072] FIG. **1** shows a box **20** arranged on a pick-up **30** using the fastening assembly **10** of present disclosure. A cross section showing the details of the fastening assembly **10** is shown in FIG. **2**.

[0073] The first aspect of this disclosure shows a fastening assembly for use in a carrier **30** such as a pick-up/truck bed comprising: one or more casing profile list **1** for being attached to an object **20** to be detachably fastened, the casing profile list **1** is arranged at the side of the object **20** facing a casing profile locking assembly arranged on the carrier **30**, the casing profile locking assembly comprising: a horizontal bottom side profile list **3**,**3'**,**35'**, one or more of a first fastening means for holding the horizontal bottom side profile list **3**,**3'**,**35'**, wherein the horizontal bottom side profile list **3**,**3'**,**35'** being fastened to the carrier **30**, and being arranged to receive the casing profile list **1**, and a vertical bottom side profile list **2**, **35"** for being arranged to interact with the horizontal bottom side profile list **3**,**3'**,**35'** for fastening the casing profile list **1** in a locking grip.

[0074] As shown in the figures the casing profile list **1** is arranged on the lower side corner of the box **20** to be detachably fastened to the top end of the pick-up truck side panels **31**. In some embodiments the casing profile list **1** is designed to be an integral part of the box itself, and then having optional receiving openings **73**, **74** for holding an end portion of a box floor **73'** or box side wall **74'**, as illustrated in FIGS. **2** and **7**. The casing profile list **1** does however not necessarily have these receiving openings **73**, **74**, as it is the outside profile form **100** that is providing for the locking grip interacting with the horizontal bottom side profile list **3**,**3'**, **35'** being fastened to the carrier **30**, and the vertical bottom side profile list **2**, **35"**. In the figures this outside form of the casing profile list **1** has an arced convex corner form and further providing an outward facing recess **71**, **72** at the end of the arc at both sides to receive corresponding protrusions **24**, **34**, **34'** of the horizontal bottom side profile list **3**,**3'**,**35'** being fastened to the carrier **30**, and the vertical bottom side profile list **2**, **35"**. The form of the recesses **71**, **72** are shown in the figures as examples only, and it should be understood that other forms may be provided within the concept of the present disclosure.

[0075] In one embodiment the casing profile list **1** thus comprise a first connecting element **71**, and the horizontal bottom side profile list **3**,**3'**,**35'** comprise a first corresponding connecting element **34**, **34'** for interacting in a guiding operation with the first connecting element **71**.

[0076] Since the horizontal bottom side profile list **3**,**3'**,**35'** is prearranged on the carrier **30**, at for example onto the top railing of the two side panels and the front panel of the pick-up bed, the first corresponding connecting element **34** of these horizontal bottom side profile list **3**,**3'**,**35'** are facing towards the object/box to be arranged on the pick-up. These first corresponding connecting element **34**, must be arranged to enable the box **20** being framed with corresponding casing profile lists **1** to enter around the span w, indicated in FIG. **5**, between the outward facing first connecting elements **34**. Thus the width between the inside of the first connecting elements **71** of the casing profile lists arranged on the sides of the box **20** must be wider than w. It is also within the concept of present disclosure to use these outward facing connecting elements **34** as guiding elements for arranging the object to be detachably fastened correctly onto the horizontal bottom side profile lists **3**,**3'**,**35'**.

[0077] In the case the fastening assembly only comprises one single horizontal bottom side profile list **3**,**3'**,**35'**, the span element becomes irrelevant. In this case the first connecting elements **71** of the casing profile list arranged on the object **20** is arranged adjacent and parallel to the outward facing connecting elements **34**, outward in the sense of the side to witch the vertical bottom side profile list **2**, **35"** is to be arranged.

[0078] The casing profile list **1** further comprises a second connecting element **72**, and the vertical bottom side profile list **2**, **35"** comprise a second corresponding connecting element **24** for interacting in a locking grip with the second connecting element **72**. When the object **20** is arranged on the horizontal bottom side profile lists **3**,**3'**,**35'**, the vertical bottom side profile list **2**, **35"** may be arranged such that the second corresponding connecting element **24** interact with the second connecting element **72**. Thus, the first corresponding connecting element **34** in combination with the second corresponding connecting element **24** interacts in a locking operation on the first connecting elements **71** and the second connecting element **72** thus holds the object **20** in a locking grip, as illustrated in FIG. **8**.

[0079] In a version adapted to be arranged in the front pointing side of the carrier the horizontal bottom side profile list **35'** and the vertical bottom side profile list **35"** may be provided in a combined bottom side profile list **35**.

[0080] The fastening assembly may further comprise two or more locking member assemblies **3'** the locking member comprises a lock member connecting element **34'** corresponding to the first corresponding connecting element **34** of the horizontal bottom side profile list **3**, and the horizontal bottom side profile list **3** comprising two or more grooves **3"** for receiving the locking members **3'**, and the locking member assembly **3'** having: at least one locking member holding means **131**, **131'**, **131"**, **131'"** for holding the locking members **3'** in the horizontal bottom side profile list groove **3"**, and at least one further locking member receiving element **132**, the horizontal bottom side profile list **3**, the further locking member receiving element **132**, and the vertical bottom side profile list **2** having corresponding further conduits **132**, **132'**, **132'"** for receiving a connecting bolt **132"** for fastening the vertical bottom side profile list **2**, **35"** to the horizontal bottom side profile list **3**,**3'**,**35'**.

[0081] One example embodiment of the locking member assembly **3'** is outlined in FIG. **3**, FIG. **4**, and FIG. **9**, wherein the locking member holding means **131**, **131'**, **131"**, **131'"** comprises a first locking member conduit **131** through the locking member. The first locking member conduit **131** runs in a direction perpendicular to the longitudinal direction of the horizontal bottom side profile list **3**,**3'**,**35'** into which the locking member assembly is to be arranged. Further a first corresponding conduit **131'** is arranged perpendicular to the longitudinal direction in the groove **3"** in the horizontal bottom side profile list **3**,**3'**,**35'**. A pin **131"**, having a diameter less than the diameter of the first locking member conduit **131**, is arranged in the combined conduits **131**, **131'** to hold the locking member safely, but able to glide on the pin **131"**, inside the groove **3"**. The pin **131"** may have threads arranged on the upper (or lower) portion towards the head, in the portion that is to reside inside the outer part of the first corresponding conduit **131'** in the horizontal bottom side profile list **3**,**3'**,**35'**. This outer (or inner) part of the first corresponding conduit **131'** in the horizontal bottom side profile list **3**,**3'**,**35'** may have corresponding threads to enable the pin **131"** to be screwed into the horizontal bottom side profile list **3**,**3'**,**35'**, and at the same time enter through the first locking member conduit **131** and the inner part of the first corresponding conduit **131'** in the horizontal bottom side profile list **3**,**3'**,**35'**. The thickness t**1** of the locking member may advantageously be less than the thickness t**2** of the groove **3"**. When t**1**<t**2** it may be room for an elastic washer member **131'"**, such as a helical coil spring washer, to be arranged on the pin **131"** at the pin **131"** head side of the locking member inside the groove **3"**, and the outer diameter of the elastic washer member **131'"** is larger than the first corresponding conduit **131'**, it will have the effect that the locking member is held resiliently (biased) towards the inner side of the groove **31"**.

[0082] A further locking member receiving element **132**, conduit, is arranged in the locking member. The further locking member conduit **132** runs in a direction perpendicular to the longitudinal direction of the horizontal bottom side profile list **3**,**3'**,**35'** into which the locking member assembly is to be arranged. A further corresponding conduit **132'** is arranged perpendicular to the longitudinal direction in the horizontal bottom side profile list **3**,**3'**,**35'**. A connecting bolt/further pin **132"**, having a diameter equal to the diameter of the further locking member conduit **132**, is arranged in the combined conduits **132**, **132'** to move the locking inside the groove **32"**. This may be achieved by providing corresponding threads on the outer portion of the connecting bolt **132"** and inside the further receiving element **132** in the locking member. The further corresponding conduit **132'** diameter is advantageously larger than the diameter of the connecting bolt **132"**. When the connecting bolt **132"** enters into the further corresponding conduit **132'** and into the locking member and is screwed it will have the effect of the locking element is moved relative the outer side of the groove **32"**.

[0083] When the object **20** is arrange onto the horizontal bottom side profile list **3**,**3'**,**35'**, and the vertical bottom side profile list **2** receiving the connecting bolt **132"** entering into the corresponding conduits **132**, **132'**, **132'"** for fastening the vertical bottom side profile list **2**, **35"** to the horizontal bottom side profile list **3**,**3'**,**35'** the object **20** is locked in the grip of the outward facing connecting elements **34** working in combination with the second corresponding connecting element **24** that interacts in a locking operation on the first connecting elements **71** and the second connecting element **72**.

[0084] The outer portion of the corresponding conduit **132'"** inside the vertical bottom side profile list **2**, **35"** may be formed in two sections, a first outer section adapted to receive the connecting bolt **132"** head, and an inner section adapted to receive the connecting bolt **132"** stem.

[0085] Thus, an even further secure locking is ensured when the connecting bolt **132"** is screwed into the locking member and pulling this towards the casing profile list **1** a distance dw, and thus the lock member connecting element **34'** is entering further dw into the corresponding first connecting elements **71** of the casing profile list **1**. In FIG. **16** it is shown the resulting forces now working to hold the fastening assembly, pushing the vertical bottom side profile list **2**, **35"** inward **26"** and at the same time the second corresponding connecting element **24** interacts with the second connecting element **72** of the casing profile list in an oblique downward movement **24'**. This is happening at the same time that the lock member connecting element **34'** is pulled outwards **26'** and oblique upwards **24"** movement toward the corresponding first connecting elements **71** of the casing profile list **1**. These moving forces ensure that the casing profile list **1** is securely held in a pinching grip sufficient to hold the object in accordance to loading regulations.

[0086] When removing the attached object **20** this is made easy by elastic washer member **131'"** pushing the connecting element **34'** inwards in the groove **32"** when the connecting bolt/further pin **132"** is removed (unscrewed), and the vertical bottom side profile list **2**, **35"** is detached. Thus there is no elements being arranged in a locking manner with the first connecting elements **71** of the casing profile list **1**. And, the object **20** may unrestricted be removed.

[0087] The horizontal bottom side profile list **3**,**3'**,**35'** further comprises on its underside a gasket groove **36** for accommodating a gasket **11**, such as a rubber gasket illustrated in FIG. **11**. The gasket will when the horizontal bottom side profile list **3**,**3'**,**35'** is arranged on the carrier **30** tighten the gap g between the bodywork of the carrier **30** and the underside of the horizontal bottom side profile list **3**,**3'**,**35'** as illustrated in FIG. **2**.

[0088] The first fastening means as discussed above and illustrated in FIG. **6** and FIG. **10** may further comprise: a vertical downward pulling fastening element **32**, **32'** for holding the horizontal bottom side profile list **3** in the vertical direction to the carrier **30**, and a horizontal working fastening element **33**, **33'** for aligning and holding the horizontal bottom side profile list **3** in the horizontal direction relative the carrier **30**.

[0089] The vertical downward pulling fastening element **32**, **32'** is comprised of: a vertical holding element **32'** being arranged in a second fastening means **50**, and a first receiving groove **32** being arranged longitudinally in the underside of the horizontal bottom side profile list **3**, the first receiving groove **32** having a form **32"** to receive and hold the vertical holding element **32'**.

[0090] The horizontal working fastening element **33**, **33'** is comprised of: a horizontal holding element **33'** being arranged in a second fastening means **50**, and a second receiving groove **33** being arranged longitudinally in a first side of the horizontal bottom side profile list **3**, the second receiving groove **33** having a form **33"** to receive and hold the horizontal holding element **33'**.

[0091] It is also provided a corner connecting element **57** for being mounted to secure perpendicular laying horizontal bottom side profile lists **3**,**3'**,**35'** to each other. Downward pulling elements **32"** is arranged to cooperate with the first receiving groove **32** of the horizontal bottom side profile list **3**,**3'**,**35'** close to the corner, and horizontal holding element **33"** is arranged to cooperate with the second receiving groove **33** of the horizontal bottom side profile list **3**,**3'**,**35'** close to the corner.

[0092] The downward pulling elements **32'**, **32"** and horizontal holding element **33'**, **33"** may be bolt and nut having a bolt head fitting into the corresponding first and second receiving groove **32**, **33**.

[0093] The horizontal bottom side profile list **3**,**3'**,**35'** may further comprise a first weight transfer/alignment element **35**, and the vertical bottom side profile list **2**, **35"** further comprising a second weight transfer/alignment element **25**, wherein the first weight transfer/alignment element **35** and the second weight transfer/alignment element **25** cooperates to transfer and hold the weight/alignment of the object **20** attached to the carrier **30**.

[0094] The second fastening means **50** may further comprise: the vertical holding element **32'**, the horizontal holding element **33'**, and a carrier fastening element **51**, **52**, **53**, **53'**, **53"** for fastening the second fastening means **50** to the carrier as illustrated in FIG. **10** and FIG. **14**.

[0095] If the carrier **30** is a pick-up truck/lorry bed having the carrier fastening elements **51**, **52**, **53**, **53'**, **53"** for fastening the second fastening means **50** to the carrier arranged on any of the top side of side and front bed panel **55**, **55'**, the topside of the panel **55**, **55'** may have an inward facing collar to provide an upside down oriented fold **55'**, and the carrier fastening elements **51**, **52**, **53**, **53'**, **53"** comprises:

- **[0096]** a lower vertical portion of an inner element **53** for being arranged on the inside of the upside down oriented fold **55'**,
- **[0097]** a lower vertical portion of an outer element **53'** being arranged on the outside of the upside down oriented fold **55'**, and
- **[0098]** a fastening device **52** arranged to pinch and hold the upside down oriented fold **55'** between the lower vertical portion of an inner element **53** and the lower vertical portion of an outer element **53'**,
- **[0099]** the inner element **53** further comprising:
- **[0100]** an upper horizontal portion for holding the vertical holding element **32'**, and
- **[0101]** an upper extension portion **53"** for holding the horizontal holding element **33'**, and the outer element **53'** further comprising:
- **[0102]** an upper horizontal portion for holding an adjustable stabilizing bolt **51**.

[0103] When mounted, the carrier fastening elements **51**, **52**, **53**, **53'**, **53"** is thus able to provide a secure non-intrusive arrangement and fastening brackets for the horizontal bottom side profile list **3**,**3'**,**35'**. The adjustable stabilizing bolt **51** provides a height adjustment feature of the horizontal bottom side profile list **3**,**3'**,**35'**, such that the object **20** may be arranged less than the gasket **11** height above the bodywork of the side panels **31** of the carrier.

[0104] The second fastening means **50** further comprise a third fastening means for further securing the second fastening means **50** to the carrier **30** such that a strap fastening element **56** providing fastening of a securing connector/strap to the carrier **30**.

[0105] In the embodiment shown in the figures the carrier **30** is a pick-up truck/lorry bed having a horizontal bottom side profile list **3**,**3'**,**35'** arranged on the top side of side and front bed panel, and the object **20** is a container having mounted. On the underside of the container **20** two longitudinal parallel running casing profile list **1** is arranged for cooperating with the horizontal bottom side profile list **3**, **3'** arranged on the top of the side panel **31** of the pick-up truck/lorry, and on the forward facing underside of the container **20** a further profile list **1** for being cooperating with the horizontal bottom side profile list **35'** that is arranged on the front panel. The second fastening means **50** are arranged on the top side of side bed panel **55**, **55'** of the pick-up truck/lorry bed.

[0106] In a further embodiment of present disclosure the first connecting element **71**, and the first corresponding connecting element **34**, **34'** of the horizontal bottom side profile list **3**,**3'**,**35'** is composed of a protrusion interacting with a groove.

[0107] In a further embodiment of present disclosure the second connecting element **72**, and the second corresponding connecting element **24** of the vertical bottom side profile list **2**, **35"** is composed of a protrusion interacting with a groove.

[0108] In the figures the first and second connecting element **71**, **72** provides the grooves of the interaction with the first and second corresponding connecting element **24**, **34**, **34'** providing the protrusions. It is within the scope of the present disclosure to switch around on this configuration such that any of the interactions may have opposite groove-protrusion configuration.

[0109] In an even further embodiment of present disclosure it is provided a roof casing profile list **1'** for being attached to an object **20** to be detachably fastened, the roof casing profile list **1'** may be arranged at a different side of the object **20** which is facing the casing profile locking assembly arranged on the carrier **30**. A further horizontal top side profile list **103**, and a vertical top side profile list **102** is provided for being arranged to interact with the horizontal top side profile list **103** for gripping the roof casing profile list **1** in a locking grip.

[0110] It is further provided a fourth fastening means **24** for securing a further object **20'** to the roof casing profile list **1'**.

[0111] It is further provided a blind casing profile list **101** for being attached to an object **20** to be detachably fastened, the blind casing profile list **101** is arranged at the lower end of the rear side of the object **20** facing the upper end of a tailgate, and a tail gasket **80** having an inner form **74**, **84** mirroring and adapted to being arranged to interact with the first and second connecting elements **71**, **72** of the blind casing profile list **101** in a locking manner, and the tail gasket **80** further the fastening comprises a downward pointing flange portion **94** being arranged to close a gap between the tail gasket and the top of the tailgate.

[0112] The second aspect of this disclosure shows a method for fastening an object to a carrier such as a pick-up/truck bed, wherein the method comprising the steps:

- **[0113]** arranging one or more casing profile list **1** according to the present disclosure to an object **20** to be arranged on a carrier **30**,
- **[0114]** mounting two or more carrier fastening element **51**, **52**, **53**, **53'**, **53"** to the carrier **30**,
- **[0115]** mounting one or more horizontal bottom side profile lists **3**,**3'**,**35'** to the two or more carrier fastening element **51**, **52**, **53**, **53'**, **53"**,
- **[0116]** mounting the casing profile list **1** arranged on the object **20** into the corresponding horizontal bottom side profile lists **3**,**3'**,**35'**,
- **[0117]** mounting one or more vertical bottom side profile list **2**, **35"** to the horizontal bottom side profile lists **3**,**3'**,**35'**, and
- **[0118]** thereby fastening the object **29** in a tight grip.

[0119] The method may comprise the further steps:

- **[0120]** arranging a blind casing profile list **1** at the lower end of the rear side of the object **20** facing the upper end of a tailgate of the carrier **30**.

[0121] The method may comprise the further steps:

- **[0122]** fastening one or more roof casing profile list **1'** to the upper portion of an object **20** at the opposite side of the object **20** facing the casing profile locking assembly arranged on the carrier **30**, and

[0123] arranging a fourth fastening means **24** for securing a further object **20'** to the roof casing profile list **1'**.

[0124] The person skilled in the art realizes that the present disclosure is not limited to the preferred embodiments described above. The person skilled in the art further realizes that modifications and variations are possible within the scope of the appended claims. Additionally, variations to the disclosed embodiments can be understood and effected by the skilled person in practicing the claimed disclosure, from a study of the drawings, the disclosure, and the appended claims.

1. A fastening assembly for use in a carrier such as a pick-up/truck bed comprising:

one or more casing profile lists attached to an object to be detachably fastened, the one or more casing profile lists arranged at a side of the object facing a casing profile locking assembly arranged on the carrier, and

the casing profile locking assembly comprising:

a horizontal bottom side profile list; and

one or more of a first fastener for holding the horizontal bottom side profile list, wherein

the horizontal bottom side profile list is fastened to the carrier, and being arranged to receive the one or more casing profile lists, and

a vertical bottom side profile list is arranged to interact with the horizontal bottom side profile list for fastening the one or more casing profile lists in a locking grip.

2. The fastening assembly according to claim **1**, wherein

the one or more casing profile lists comprise a first connecting element, and

the horizontal bottom side profile list comprises a first corresponding connecting element for interacting in a guiding operation with the first connecting element.

3. The fastening assembly according to claim **1**, wherein

the one or more casing profile lists comprise a second connecting element, and

the vertical bottom side profile list comprises a second corresponding connecting element for interacting in a locking grip with the second connecting element.

4. The fastening assembly according to claim **1**, wherein

the horizontal bottom side profile list and the vertical bottom side profile list are provided in a combined bottom side profile list.

5. The fastening assembly according to claim **2**, further comprising:

two or more locking member assemblies comprising a lock member connecting element corresponding to the first corresponding connecting element of the horizontal bottom side profile list,

the horizontal bottom side profile list comprising two or more grooves for receiving the two or more locking member assemblies,

a locking member assembly of the two or more locking member assemblies comprising:

at least one locking member holder for holding the locking member assembly in a horizontal bottom side profile list groove of the two or more grooves; and

at least one locking member receiving element, and

the horizontal bottom side profile list, the at least one locking member receiving element, and the vertical bottom side profile list comprising corresponding conduits for receiving a connecting bolt for fastening the vertical bottom side profile list to the horizontal bottom side profile list.

6. The fastening assembly according to claim **1**, wherein the first fastener comprising:

a vertical downward pulling fastening element for holding the horizontal bottom side profile list in a vertical direction to the carrier; and

a horizontal working fastening element for aligning and holding the horizontal bottom side profile list in a horizontal direction relative the carrier.

7. The fastening assembly according to claim **6**, wherein the vertical downward pulling fastening element comprises:

a vertical holding element arranged in a second fastener; and

a first receiving groove arranged longitudinally in an underside of the horizontal bottom side profile list, the first receiving groove having a form to receive and hold the vertical holding element.

8. The fastening assembly according to claim **6**, wherein the horizontal working fastening element comprises:

a horizontal holding element arranged in a second fastener; and

a second receiving groove arranged longitudinally in a first side of the horizontal bottom side profile list, the second receiving groove having a form to receive and hold the horizontal holding element.

9. The fastening assembly according to claim **1**, wherein

the horizontal bottom side profile list further comprises a first weight transfer/alignment element,

the vertical bottom side profile list further comprises a second weight transfer/alignment element, and

the first weight transfer/alignment element and the second weight transfer/alignment element cooperates to transfer and hold the weight of, and/or align, the object attached to the carrier.

10. The fastening assembly according to claim **7**, wherein the second fastener comprises:

the vertical holding element;

a horizontal holding element arranged in the second fastener; and

a carrier fastening element for fastening the second fastener to the carrier.

11. The fastening assembly according to claim **10**, wherein

the carrier is a pick-up truck/lorry bed having carrier fastening elements for fastening the second fastener to the carrier arranged on any of a top side of side and front bed panel, a topside of the side and front bed panel having an inward facing collar to provide an upside down oriented fold,

the carrier fastening elements comprises:

a lower vertical portion of an inner element arranged on an inside of the upside down oriented fold;

a lower vertical portion of an outer element or arranged on an outside of the upside down oriented fold; and

a fastening device arranged to pinch and hold the upside down oriented fold between the lower vertical portion of the inner element and the lower vertical portion of the outer element,

the inner element further comprises:

an upper horizontal portion for holding the vertical holding element; and an upper extension portion for holding the horizontal holding element, and

the outer element further comprises:

an upper horizontal portion for holding a stabilizing bolt.

12. The fastening assembly according to claim **7**, wherein the second fastener further comprises:

a third fastener for further securing the second fastener to the carrier.

13. The fastening assembly according to claim **12**, wherein the third fastener comprises a strap fastening element configured to fasten a securing connector/strap to the carrier.

14. The fastening assembly according to claim **1**, wherein

the carrier is a pick-up truck/lorry bed having the horizontal bottom side profile list arranged on a top side of side and front bed panel, and the object is a container having mounted:

on an underside of the object, two longitudinal parallel running casing profile lists for cooperating with the horizontal bottom side profile list arranged on a side panel, and

on a forward facing underside of the object, a further profile list for cooperating with the horizontal bottom side profile list arranged on a front panel, and

a second fastener is arranged on a top side of a side bed panel of the side and front bed panel of the pick-up truck/lorry bed.

15. The fastening assembly according to claim **2**, wherein the first connecting element and the first corresponding connecting element of the horizontal bottom side profile list comprise a protrusion interacting with a groove.

16. The fastening assembly according to claim **3**, wherein the second connecting element, and the second corresponding connecting element of the vertical bottom side profile list comprise a protrusion interacting with a groove.

17. The fastening assembly according to claim **1**, further comprising:

one or more roof casing profile lists attached to the object to be detachably fastened, the one or more roof casing profile lists arranged at a different side of the object facing the casing profile locking assembly arranged on the carrier;

a horizontal top side profile list; and

a vertical top side profile list arranged to interact with the horizontal top side profile list for gripping the one or more roof casing profile lists in a locking grip.

18. The fastening assembly according to claim **17**, further comprising:

a fourth fastener for securing a further object to the one or more roof casing profile lists.

19. The fastening assembly according to claim **1**, further comprising:

a blind casing profile list attached to an object to be detachably fastened, the blind casing profile list is arranged at a lower end of a rear side of the object facing an upper end of a tailgate; and

a tail gasket having an inner form mirroring and adapted to be arranged to interact with first and second connecting elements of the blind casing profile list in a locking manner, and the tail gasket further comprising a downward pointing flange portion arranged to close a gap between the tail gasket and a top of the tailgate.

20. A method for fastening an object to a carrier such as a pick-up/truck bed, the method comprising:

arranging the one or more casing profile lists according to claim **1** to the object to be arranged on a carrier;

mounting two or more carrier fastening elements to the carrier;

mounting one or more horizontal bottom side profile lists to the two or more carrier fastening elements;

mounting the one or more casing profile lists arranged on the object into corresponding one or more horizontal bottom side profile lists; and

mounting one or more vertical bottom side profile list to the one or more horizontal bottom side profile lists, thereby fastening the object in a tight grip.

21. The method according to claim **20**, further comprising:

arranging a blind casing profile list at a lower end of a rear side of the object facing an upper end of a tailgate of the carrier.

22. The method according to claim **20**, further comprising:

fastening one or more roof casing profile lists to an upper portion of the object at an opposite side of the object facing the casing profile locking assembly arranged on the carrier; and

arranging a fourth fastener for securing a further object to the one or more roof casing profile lists.

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